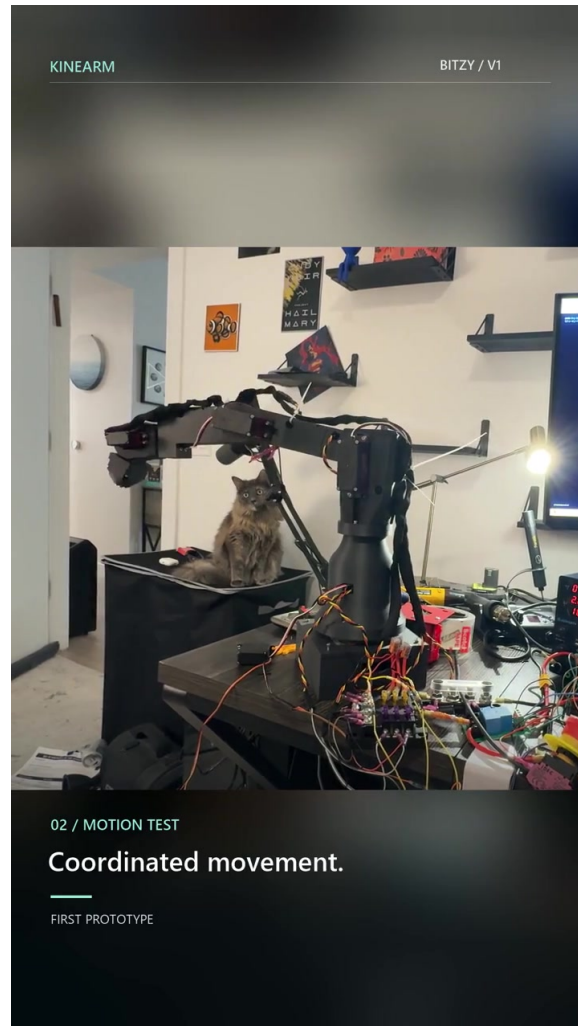


## The built five-axis prototype

The physical demo shows the printed arm, paired shoulder architecture, exposed prototype electronics and a removable tool module. It provides visual assembly and movement evidence.



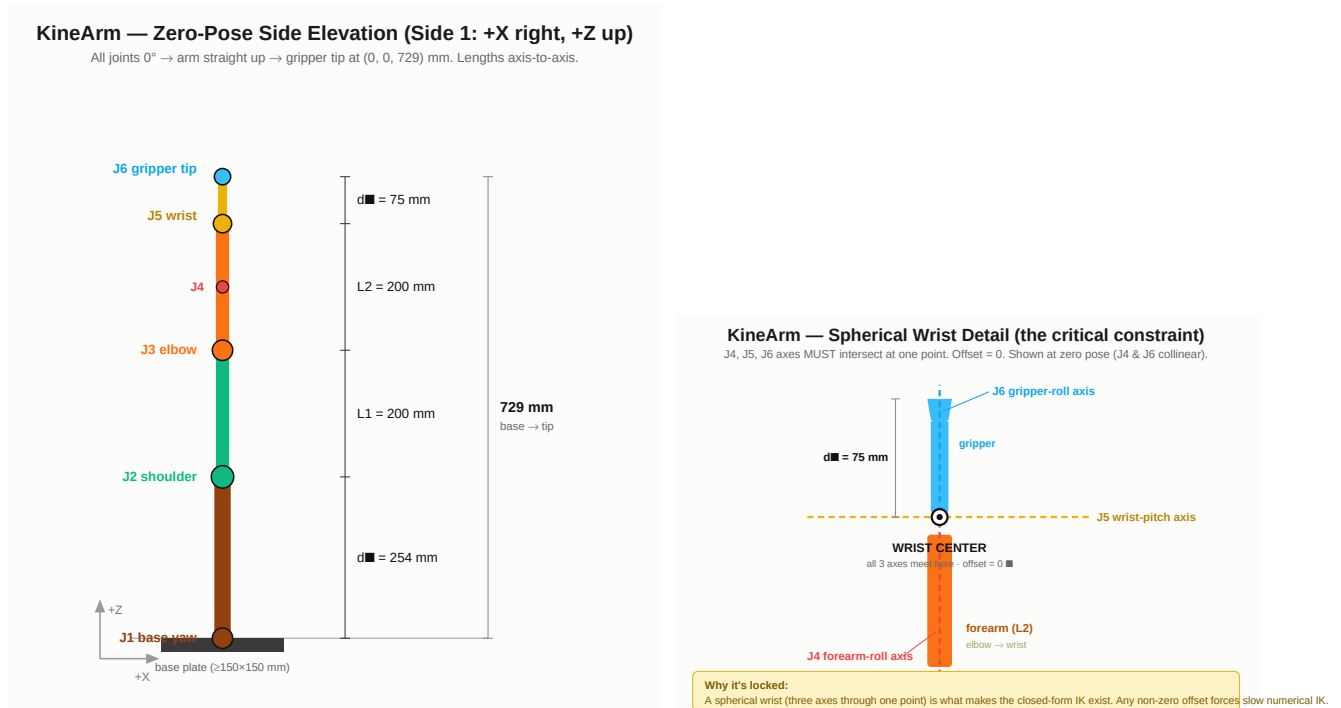
**Manual module exchange.** Seat the original star interface and retain the module with the four mounting screws. The recorded prototype sequence is manual; it is not automatic tool changing.

The CAD tool revisions retain the original mounting interface. Their physical fit and installed revision must be checked separately. The demo does not measure positioning, payload or repeatability.

AI-assisted workflows supported the creation of this project.

## Earlier six-axis design references

These original sketches describe the earlier six-axis study. The 729 mm zero-pose reference and original servo labels belong to that revision. They do not certify the current five-axis arm dimensions or hardware identities.



The sketch progression explains link and joint relationships used during kinematic development. Current commissioned limits and actual installed geometry govern physical work.

The present nominal reach reference is approximately 475 mm. The 500 g object-plus-tool objective remains a target. A full as-built dimensional survey and measured load envelope are not included in this release.

Review composition from existing sketches and prototype footage; no editable engineering sources supplied.